

MRC Lecture 4 — Derivation Guide 1

The Boltzmann Distribution from Maximum Entropy

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1. Motivation

The **Boltzmann distribution** is the governing equation of molecular recognition at thermal equilibrium. Every binding measurement ever made — by SPR, ITC, BLI, or any other technique — is a measurement of this distribution. In Lectures 1–3, students encountered the same function as **softmax** in the transformer attention mechanism. This guide derives the Boltzmann distribution from first principles using **maximum entropy** and **Lagrange multipliers**, proving that it is the most probable distribution of a system at thermal equilibrium.

The derivation was first given by Ludwig Boltzmann in 1877 and reformulated by Josiah Willard Gibbs in the early 1900s. Edwin Jaynes (1957) showed that the same derivation follows from information theory: the Boltzmann distribution is the **maximum entropy** distribution — the probability assignment that is maximally noncommittal given only a constraint on the average energy. This guide follows the Boltzmann-Jaynes approach.

2. Setup: The Statistical Mechanics Problem

Physical system. Consider N particles (or molecules) distributed among M energy states with energies $E_1, E_2, \dots, E_M$. Let n_j denote the number of particles in state j , called the **occupation number**.

Two constraints:

(1) **Conservation of particles.** The total number of particles is fixed:

$$\sum_{j=1}^M n_j = N$$

(2) **Conservation of energy.** The total energy of the system is fixed:

$$\sum_{j=1}^M n_j E_j = U$$

where U is the total energy of the system.

The question. Subject to these two constraints, what distribution of occupation numbers $\{n_1, n_2, \dots, n_M\}$ is **most probable** — i.e., corresponds to the largest number of microscopic arrangements (microstates)?

3. The Multiplicity

Definition. The number of distinct ways to arrange N distinguishable particles into groups of sizes $n_1, n_2, \dots, n_M$ is the **multinomial coefficient**:

$$W = \frac{N!}{n_1! n_2! \cdots n_M!} = \frac{N!}{\prod_{j=1}^M n_j!}$$

W counts the number of **microstates** consistent with the macrostate $\{n_1, \dots, n_M\}$. The most probable macrostate is the one that maximizes W .

Why maximize W ? At equilibrium, a thermodynamic system has had time to explore all accessible microstates. The macrostate with the most microstates is the one the system is most likely to be found in — by an overwhelmingly large factor for macroscopic systems ($N \sim 10^{23}$).

Connection to entropy. Boltzmann's entropy is:

$$S = k_B \ln W$$

Maximizing W is equivalent to maximizing the entropy S . This is the statistical mechanics statement of the second law: at equilibrium, entropy is maximized.

4. Stirling's Approximation

For large n , the factorial is well approximated by **Stirling's formula**:

$$\ln(n!) \approx n \ln n - n$$

This approximation is excellent for $n \gg 1$ (the relative error is $O(1/n)$) and is standard in statistical mechanics.

Applying Stirling's approximation to $\ln W$:

$$\begin{aligned} \ln W &= \ln N! - \sum_{j=1}^M \ln(n_j!) \\ &\approx (N \ln N - N) - \sum_{j=1}^M (n_j \ln n_j - n_j) \\ &= N \ln N - N - \sum_{j=1}^M n_j \ln n_j + \sum_{j=1}^M n_j \end{aligned}$$

Since $\sum n_j = N$, the $-N$ and $+\sum n_j$ cancel:

$$\ln W \approx N \ln N - \sum_{j=1}^M n_j \ln n_j$$

This can be rewritten in terms of probabilities $P_j = n_j / N$:

$$\begin{aligned} \ln W &\approx N \ln N - \sum_{j=1}^M N P_j \ln(N P_j) = N \ln N - N \sum_{j=1}^M P_j (\ln N + \ln P_j) \\ &= N \ln N - N \ln N \sum_{j=1}^M P_j - N \sum_{j=1}^M P_j \ln P_j \end{aligned}$$

Since $\sum P_j = 1$:

$$\ln W \approx -N \sum_{j=1}^M P_j \ln P_j$$

The quantity $H = -\sum P_j \ln P_j$ is the **Shannon entropy** (Shannon, 1948). Maximizing W is equivalent to maximizing the Shannon entropy H — the connection Jaynes (1957) used to unify statistical mechanics and information theory.

5. Constrained Optimization with Lagrange Multipliers

The optimization problem. Maximize $\ln W$ (equivalently, the Shannon entropy H) subject to the two constraints. We use the method of Lagrange multipliers.

The Lagrangian. Introduce multipliers α for particle conservation and β for energy conservation:

$$L = \ln W - \alpha \left(\sum_{j=1}^M n_j - N \right) - \beta \left(\sum_{j=1}^M n_j E_j - U \right)$$

Substituting the Stirling approximation for $\ln W$:

$$L = N \ln N - \sum_{j=1}^M n_j \ln n_j - \alpha \left(\sum_{j=1}^M n_j - N \right) - \beta \left(\sum_{j=1}^M n_j E_j - U \right)$$

Take the derivative with respect to n_j and set to zero:

$$\frac{\partial L}{\partial n_j} = -\ln n_j - 1 - \alpha - \beta E_j = 0$$

(Note: $\partial(n_j \ln n_j) / \partial n_j = \ln n_j + 1$ by the product rule.)

Solve for n_j :

$$\begin{aligned} \ln n_j &= -1 - \alpha - \beta E_j \\ n_j &= \exp(-1 - \alpha) \cdot \exp(-\beta E_j) \end{aligned}$$

Define $C = \exp(-1 - \alpha)$, which is a constant (independent of j):

$$n_j = C \cdot \exp(-\beta E_j)$$

6. Normalization: Determining C

The constant C is determined by the particle conservation constraint $\sum n_j = N$:

$$\begin{aligned} \sum_{j=1}^M n_j &= C \sum_{j=1}^M \exp(-\beta E_j) = N \\ C &= \frac{N}{\sum_{j=1}^M \exp(-\beta E_j)} \end{aligned}$$

The probability of being in state j is:

$$P_j = \frac{n_j}{N} = \frac{C \cdot \exp(-\beta E_j)}{N} = \frac{\exp(-\beta E_j)}{\sum_{k=1}^M \exp(-\beta E_k)}$$

7. Identifying β : The Temperature

The Lagrange multiplier β is identified with the inverse temperature through the thermodynamic relation:

$$\beta = \frac{1}{k_B T}$$

where $k_B = 1.381 \times 10^{-23}$ J/K is **Boltzmann's constant** and T is the absolute temperature in Kelvin.

This identification comes from requiring consistency with the fundamental thermodynamic relation $dS = dU / T$ at equilibrium. The derivation of this identification is a standard result in statistical mechanics (see Dill & Bromberg, Chapter 10, for the complete argument).

For molecular applications, it is conventional to use the gas constant $R = N_A k_B = 1.987 \times 10^{-3}$ kcal/(mol·K), which gives energies in kcal/mol rather than per-molecule units. Then $\beta = 1 / RT$.

8. The Boltzmann Distribution: Final Result

Substituting $\beta = 1 / k_B T$:

$$P(j) = \frac{\exp(-E_j / k_B T)}{\sum_{k=1}^M \exp(-E_k / k_B T)}$$

This is the **Boltzmann distribution**. The denominator:

$$Z = \sum_{k=1}^M \exp(-E_k / k_B T)$$

is the **partition function** (German: Zustandssumme, “sum over states”).

9. Properties of the Boltzmann Distribution

Property 1: Positivity. $P(j) > 0$ for all j , since the exponential is always positive. Every state has non-zero probability at non-zero temperature. No state is ever completely excluded.

Property 2: Normalization. $\sum_{j=1}^M P(j) = 1$ by construction (the denominator ensures this).

Property 3: Lower energy $\rightarrow$ higher probability. If $E_A < E_B$ (state A has lower energy), then $\exp(-E_A / k_B T) > \exp(-E_B / k_B T)$, so $P(A) > P(B)$. The distribution favors lower-energy states.

Property 4: Temperature controls sharpness.

- As $T \rightarrow 0$ (low temperature): $P(j) \rightarrow 1$ for the lowest-energy state and $P(j) \rightarrow 0$ for all others. The system locks into its ground state. This is the **argmax** limit.
- As $T \rightarrow \infty$ (high temperature): $\exp(-E_j/k_B T) \rightarrow 1$ for all j , so $P(j) \rightarrow 1/M$ for all j . The distribution becomes uniform. All states are equally likely.
- At intermediate temperature: the distribution is graded — lower-energy states are preferred but higher-energy states retain non-zero probability.

These are exactly the three temperature regimes from Lecture 1 (Derivation Guide 5).

Property 5: Shift invariance. Adding a constant C to all energies does not change the distribution:

$$\frac{\exp(-(E_j + C)/k_B T)}{\sum_k \exp(-(E_k + C)/k_B T)} = \frac{\exp(-C/k_B T) \exp(-E_j/k_B T)}{\exp(-C/k_B T) \sum_k \exp(-E_k/k_B T)} = \frac{\exp(-E_j/k_B T)}{\sum_k \exp(-E_k/k_B T)}$$

Only energy **differences** matter — the absolute energy scale is arbitrary. This is the same shift invariance property of softmax derived in Lecture 1 (Derivation Guide 4, §8).

10. The Boltzmann Distribution IS Softmax

Claim. The Boltzmann distribution is mathematically identical to the softmax function.

Proof. Define the dimensionless score:

$$s_j = -\frac{E_j}{k_B T}$$

This is the negated energy in units of the thermal energy $k_B T$. Higher score = lower energy = more favorable.

Substituting into the Boltzmann distribution:

$$P(j) = \frac{\exp(-E_j/k_B T)}{\sum_k \exp(-E_k/k_B T)} = \frac{\exp(s_j)}{\sum_k \exp(s_k)} = \text{softmax}(s)_j \blacksquare$$

The Boltzmann distribution is softmax applied to the dimensionless energy scores. The partition function Z is the softmax denominator. The inverse temperature $\beta = 1/k_B T$ is the temperature parameter $1/\tau$.

The correspondence table:

Boltzmann distribution	Softmax / Transformers	Physical meaning
Energy E_j	Score $-z_j$ (negated)	How unfavorable state j is
Inverse temperature $\beta = 1/k_B T$	Inverse temperature $1/\tau$	Selection sharpness
Partition function Z	Denominator $\sum \exp(z_k)$	Normalizer
Occupation probability $P(j)$	Softmax output	Selection probability
Maximum entropy	Maximum entropy	Least biased assignment

11. Worked Example: Three Energy States

Problem. A system has three states with energies $E_1 = -2$, $E_2 = 0$, $E_3 = 3$ (in units of $k_B T$, so we work with dimensionless energies and $\beta = 1$). Compute the Boltzmann probabilities.

Step 1. Compute scores $s_j = -E_j / k_B T = -E_j$ (since $k_B T = 1$):

$$s_1 = -(-2) = 2, s_2 = -(0) = 0, s_3 = -(3) = -3$$

Step 2. Exponentiate:

$$\exp(s_1) = \exp(2) = 7.389, \exp(s_2) = \exp(0) = 1.000, \exp(s_3) = \exp(-3) = 0.050$$

Step 3. Partition function:

$$Z = 7.389 + 1.000 + 0.050 = 8.439$$

Step 4. Probabilities:

$$P(1) = \frac{7.389}{8.439} = 0.876, P(2) = \frac{1.000}{8.439} = 0.119, P(3) = \frac{0.050}{8.439} = 0.006$$

Verify: $0.876 + 0.119 + 0.006 = 1.001 \approx 1.000$ ✓ (rounding)

Interpretation. State 1 (lowest energy, $E_1 = -2$) receives 87.6% of the probability. State 2 (zero energy) receives 11.9%. State 3 (highest energy, $E_3 = 3$) receives only 0.6%. The Boltzmann distribution strongly favors the lowest-energy state, but does not exclude the others.

This is the same computation as the softmax examples in Lecture 1 — the scores are $[2, 0, -3]$ and the probabilities are $[0.876, 0.119, 0.006]$.

12. Worked Example: Effect of Temperature

Problem. Using the same three states ($E_1 = -2$, $E_2 = 0$, $E_3 = 3$ kcal/mol), compute the Boltzmann probabilities at three temperatures: $T = 155$ K ($k_B T = 0.308$ kcal/mol), $T = 310$ K ($k_B T = 0.616$ kcal/mol), and $T = 620$ K ($k_B T = 1.232$ kcal/mol).

At $T = 155$ K ($k_B T = 0.308$ kcal/mol):

$$s_1 = 2 / 0.308 = 6.49, s_2 = 0 / 0.308 = 0, s_3 = -3 / 0.308 = -9.74$$

$$\exp(6.49) = 659.9, \exp(0) = 1.000, \exp(-9.74) = 0.000059$$

$$Z = 660.9, P(1) = 0.999, P(2) = 0.002, P(3) \approx 0.000$$

Nearly deterministic — State 1 receives essentially all probability.

At $T = 310$ K ($k_B T = 0.616$ kcal/mol):

$$s_1 = 2 / 0.616 = 3.25, s_2 = 0, s_3 = -3 / 0.616 = -4.87$$

$$\exp(3.25) = 25.79, \exp(0) = 1.000, \exp(-4.87) = 0.0077$$

$$Z = 26.80, P(1) = 0.962, P(2) = 0.037, P(3) = 0.000$$

State 1 strongly favored, but State 2 has meaningful probability.

At $T = 620$ K ($k_B T = 1.232$ kcal/mol):

$$s_1 = 2 / 1.232 = 1.62, s_2 = 0, s_3 = -3 / 1.232 = -2.44$$

$$\exp(1.62)=5.053, \exp(0)=1.000, \exp(-2.44)=0.087$$

$$Z=6.140, P(1)=0.823, P(2)=0.163, P(3)=0.014$$

More diffuse — State 2 and even State 3 have non-negligible probability.

Summary:

Temperature	$k_B T$	$P(E = -2)$	$P(E = 0)$	$P(E = 3)$	Character
155 K	0.308	0.999	0.002	0.000	Nearly deterministic
310 K	0.616	0.962	0.037	0.000	Strongly selective
620 K	1.232	0.823	0.163	0.014	More diffuse

Higher temperature → flatter distribution → less selective. This is exactly the temperature effect from Lecture 1 (Derivation Guide 5), now derived from thermodynamics.

13. The Jaynes Perspective: Maximum Entropy as Rational Inference

Jaynes (1957) reframed the Boltzmann derivation as an inference problem rather than a counting problem:

Given: The average energy $\langle E \rangle = \sum P_j E_j$ is known (from measurement or constraint).

Sought: The probability distribution $\{P_1, \dots, P_M\}$ that makes the fewest assumptions beyond this constraint.

Method: Maximize the Shannon entropy $H = -\sum P_j \ln P_j$ subject to $\sum P_j = 1$ and $\sum P_j E_j = \langle E \rangle$.

Result: The Boltzmann distribution — exactly the same function.

Jaynes' insight is that the Boltzmann distribution is not just physics — it is **epistemology**. It is the uniquely rational probability assignment when you know the average energy and nothing else. Any other distribution would smuggle in additional assumptions not justified by the available information. This connects directly to the Luce uniqueness theorem (Lecture 6): Luce showed that the exponential form is the unique function satisfying five axioms of rational choice. Jaynes showed it is the unique function satisfying the maximum entropy principle. Both arrive at the same function from different axioms — a convergence that the MRC framework extends to molecular recognition and transformer attention.

14. Summary

Concept	Result	Significance
Multiplicity	$W = N! / \prod n_j!$	Number of microstates for a macrostate
Stirling's approx.	$\ln n! \approx n \ln n - n$	Makes the optimization tractable
Shannon entropy	$H = -\sum P_j \ln P_j$	Maximizing W = maximizing H
Lagrange multipliers	α (particles), β (energy)	Enforce constraints during optimization
The key equation	$\partial L / \partial n_j = 0 \rightarrow n_j = C \exp(-\beta E_j)$	The exponential form emerges
Normalization	$P(j) = \exp(-\beta E_j) / Z$	The Boltzmann distribution

Concept	Result	Significance
Temperature	$\beta = 1 / k_B T$	Connects the multiplier to physics
Partition function	$Z = \sum \exp(-E_k / k_B T)$	The softmax denominator
Identity	Boltzmann = softmax with $s_j = -E_j / k_B T$	The equation that governs both physics and AI

15. Checkpoint Questions

Q1. In the derivation, we maximized $\ln W$ rather than W directly. Why is this valid?

Answer: The logarithm is a monotonically increasing function: if $\ln W_A > \ln W_B$, then $W_A > W_B$. Therefore, maximizing $\ln W$ gives the same result as maximizing W . The logarithm is preferred because it converts the product in $W = N! / \prod n_j!$ into a sum, making the calculus tractable. Additionally, Stirling's approximation applies to $\ln(n!)$, not $n!$ directly.

Q2. The derivation uses two Lagrange multipliers (α and β) for two constraints. If there were an additional constraint — say, a fixed average of some other quantity $\langle X \rangle = \sum P_j X_j$ — what would the resulting distribution look like?

Answer: Adding a third constraint introduces a third multiplier γ , and the result is $P(j) \propto \exp(-\beta E_j - \gamma X_j)$. The distribution is still exponential, but with two terms in the exponent. This is the generalized Boltzmann distribution. In the MRC framework, this corresponds to adding an additional feature to the score: $s_j = -\beta E_j - \gamma X_j$, which is a dot product of a two-dimensional feature vector. This is why the bilinear decomposition (Lecture 5) naturally generalizes: more contact features $\rightarrow$ higher-dimensional dot product.

Q3. At what temperature does the Boltzmann distribution become uniform (all states equally probable)?

Answer: As $T \rightarrow \infty$, $\beta = 1 / k_B T \rightarrow 0$, so $\exp(-\beta E_j) \rightarrow \exp(0) = 1$ for all j . The distribution becomes $P(j) = 1 / M$ for all j — uniform. Infinite temperature means the thermal energy overwhelms all energy differences, so the system has no preference among states. In softmax terms, this is $\tau \rightarrow \infty$.

Q4. A student claims: “The Boltzmann distribution only applies to gas molecules. Molecular binding is different.” How would you respond?

Answer: The Boltzmann distribution applies to any system at thermal equilibrium, not just gases. The derivation assumes only: (i) the system has discrete energy states, (ii) total particles and total energy are conserved, (iii) the system is in the most probable macrostate. Molecular binding at thermal equilibrium satisfies all three. Every K_d measurement ever made — for proteins, nucleic acids, small molecules, any molecular pair — is a measurement of the Boltzmann probability ratio between bound and unbound states. The distribution is universal for equilibrium systems.

16. Connection to Future Lectures

- **Lecture 4, WB2:** The Boltzmann distribution is applied to the two-state system (bound vs. unbound) to derive $K_d = \exp(\Delta G / RT)$ and $\Delta G = RT \ln K_d$.
- **Lecture 4, WB3:** The Boltzmann distribution is applied to competitive binding (one agent, M targets) to show that the selection probability IS softmax.
- **Lecture 5:** The energy E_j in the Boltzmann distribution decomposes as a dot product in nucleic acid systems (given bilinearity) and must be learned in protein systems (emergent bilinearity). This is the bridge from thermodynamics to the transformer architecture.
- **Lecture 6:** The convergence equation — the Boltzmann distribution = softmax — is proven to be the unique function satisfying five axioms. The maximum entropy derivation given here is one of six independent derivations of the same function.
- **Lecture 7:** The Boltzmann distribution, combined with five architecture conditions, prescribes the complete SFM architecture. The partition function becomes the softmax denominator. The Lagrange multiplier β becomes the learned inverse temperature $1 / \tau$.
- **Lecture 10:** The affinity calibration framework derives $\Delta G = ws + b$ from the identity $s_{ij} = -\Delta G_{ij} / k_B T + C$, which follows directly from the Boltzmann distribution derived here.